

The mysterious universe: Summary

Answers

Use the listed words to complete the sentences.

absolute	protostar	electrons	spiral	observer
constellations	space	giants	billion	radio
fusion	time	nuclei	expansion	temperature
nebulae	afterglow	pulsars	helium	

- Groups of stars form patterns that are calledconstellations.....
- Our solar system is part of a largespiral..... galaxy called the Milky Way.
- Clouds of dust and gas in interstellar space are callednebulae.....
- The apparent movement of stars at different distances is caused by the movement of theobserver..... This effect is called parallax.
- According to the big bang theory, the universe is about 13.7billion..... years old.
- The first stage of the big bang involved the formation oftime..... and space. The early universe was very small and very hot. Space then underwent rapidexpansion..... and as it expanded it cooled.
- Quarks andelectrons..... were the first matter particles to form. Subsequent cooling led to proton and neutron formation, which then combined to formnuclei.....
- Red shifting of lines in star spectra is consistent with the stretching ofspace..... as the universe expands.
- The big bang theory can explain the large amount ofhelium..... relative to hydrogen in the universe.
- Microwave background radiation has been detected by instruments aboard satellites. This microwave radiation is the cooledafterglow..... of the big bang.
- Stars are formed when gravity causes gas and dust to come together and heat up. Eventually aprotostar..... is formed and then finally a star of a given mass.
- Stars like our sun evolve into redgiants..... before they explode.
- Red supergiants evolve into black holes orpulsars.....
- Theabsolute..... magnitude of a star is a measure of its real brightness when all stars are compared at the same distance.
- The Hertzsprung–Russell diagram plots the surfacetemperature..... of the star versus its absolute magnitude. This diagram helps us understand the evolutionary life cycle of a star.
- Stars generate their energy by nuclearfusion..... in which hydrogen nuclei join together to form heavier nuclei.
- The universe can be studied using optical (visible light) telescopes as well asradio..... telescopes.